

AMRESHWAR SINGH

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EDUCATION

VIT Bhopal University, Bhopal, Madhya Pradesh – 466114

Bachelor of Technology in Computer Science

With specialization in Artificial Intelligence & Machine Learning

Cumulative GPA: 7.94

May 2027

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, Node.js, REST APIs, FastAPI, Google Cloud Platform

Databases & Retrieval: PostgreSQL, SQLite, FAISS, ChromaDB

AI / Agentic Systems: LangChain, HuggingFace, Vertex AI, multi-agent orchestration, parallel worker architectures

Dev & Deployment: Git, Docker, Streamlit, Linux, agent-based tooling

PROJECTS

Mental Wellness AI Agent | Multi-Agent Distributed Architecture | [GitHub](#)

- Built a low-latency crisis detection module with deterministic routing to emergency resources (988, Crisis Text Line), ensuring immediate escalation independent of conversational state.
- Architected a distributed 3-agent system (Analysis, Messaging, Orchestrator) with asynchronous message-passing, partitioned responsibilities, and well-defined inter-agent contracts.
- Designed a finite-state decision engine across 10 mental states $\times$ 4 intensity levels, dynamically generating personalized 7-day therapeutic programs from real-time user data.
- Used PostgreSQL, redis, mem0 as a persistent cross-session store running on docker engine; integrated Google Calendar and Spotify APIs across 4 automated daily workflows.

General-Purpose RAG Pipeline | Scalable Retrieval & Data Engineering | [GitHub](#)

- Engineered a parallel batch-processing system for 100+ documents with SQLite-backed crash recovery, guaranteeing exactly-once processing semantics under interruption.
- Designed a hybrid retrieval algorithm combining FAISS ANN search, ChromaDB vector storage, and a two-stage reranking layer for high-precision retrieval over large document corpora.
- Built a modular 7-stage pipeline (ingestion $\rightarrow$ chunking $\rightarrow$ embedding $\rightarrow$ indexing $\rightarrow$ retrieval $\rightarrow$ reranking $\rightarrow$ generation) architected for horizontal scalability and component-level replaceability.
- Resolved non-digital document ingestion using dual OCR engines (Tesseract + EasyOCR) with overlapping chunking to preserve semantic context at chunk boundaries.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- Applied Machine Learning in Python — University of Michigan (Coursera)
- Microsoft Certified: Azure Data Fundamentals — Microsoft
- Generative AI in Action — IBM SkillsBuild

EXTRA-CURRICULAR

Entrepreneurship Cell, VIT Bhopal

Content Lead — Operations, Finance & Media

Jan 2024 – Aug 2025

- Managed content writing and documentation across Operations, Finance, and Media; established guided entrepreneurship communities within the university.
- Authored sponsorship brochures and memos that secured over ₹50k in sponsorships.
- Led and coordinated a team of 8 members across two consecutive tenures.